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Development of Ampelopsis Radix Ethanolic Extract Loaded Phytosomes for Improved Efficacy in Colorectal Cancer: *in vitro* and *in vivo* Assessment Study

Hongze Wu[1], Bo Wang[2], Xia Li[1], Chao Lu[3], Qishu Zeng[4], Lin Lu[3], Ming Chen[1], and Yueran Wu[5]*

[1] *Department of Spleen and Stomach Diseases, Jiujiang Hospital of Chinese Medicine, Jiujiang, 332000, CHINA*
[2] *Department of Oncology, Jiujiang Hospital of Chinese Medicine, Jiujiang, 332000, CHINA*
[3] *Anorectal Department, Jiujiang Hospital of Chinese Medicine, Jiujiang, 332000, CHINA*
[4] *Medical Department, Jiujiang Hospital of Chinese Medicine, Jiujiang, 332000, CHINA*
[5] *Chronic Disease Department, Jiujiang Center for Disease Control and Prevention, Jiujiang, 332000, CHINA*

Abstract: The aim of present work was to develop and evaluate Ampelopsis Radix ethanolic extract loaded phytosomes for improved efficacy in colorectal cancer. Ampelopsis Radix ethanolic extract was prepared by Soxhlet extraction process followed by development of phytosomes using lipids and other excipients. The phytosomes were evaluated for surface morphology, particle size analysis, zeta potential, encapsulation efficiency, drug loading, *in vitro* drug release, Cytotoxicity assay, cellular uptake studies were performed on HCT-116 and SW480 cell lines. *In vivo* antitumor activity was performed. The phytosomes were found spherical shape with smooth surface characteristics. The drug loading was observed between 29.27 to 42.10 % while particle size of 85 to 130 nm was found. Phytosomes showed desired release pattern which is required for cancer treatment. Phytosomes showed maximum antiproliferative activity on cell lines over the period of 24 hours and showed highest internalization within both types of cell lines. The survival rate of animals in phytosomes treated group was found to be 100% proving the safety and efficacy. Phytosomes showed highest antitumor activity as compared to other formulations. Study confirms the potential use Ampelopsis Radix ethanolic extract loaded phytosomes for improved efficacy in colorectal cancer.

Key words: antitumor, drug loading, herbal extract, cell line, sustained release

1 Introduction

Colorectal cancer, also known as colon or rectal cancer, is a common digestive system malignancy originating from benign colon or rectum polyps that can turn cancerous. Risk factors include age, family history, and lifestyle choices[1)]. In early stages, it may be asymptomatic, but as it progresses, symptoms like bowel habit changes, blood in stools, abdominal pain, weight loss, and fatigue may appear. Early detection via screening, like colonoscopy or CT scans, is crucial. Staging guides treatment choices, which include surgery, chemotherapy, radiation, targeted therapy, and immunotherapy[2)]. Prevention involves healthy lifestyle habits, and prognosis varies by stage, with early detection improving survival rates[3)].

The treatment of colorectal cancer involves a variety of therapeutic agents, each with its own mechanisms and potential drawbacks. Chemotherapy agents such as 5-Fluorouracil (5-FU), Oxaliplatin, and Irinotecan can be effective but may lead to side effects like nausea, diarrhea, neuropathy, and myelosuppression[4)]. Targeted therapies like Cetuximab, Panitumumab, and Bevacizumab are effective in certain patients with specific genetic mutations but can cause skin rashes, gastrointestinal issues, and high blood pressure. Regorafenib may result in fatigue and hand-foot skin reactions. Immunotherapy with immune checkpoint inhibitors, such as Pembrolizumab, is well-tolerated but may lead to immune-related adverse events like colitis and hepatitis[5)]. Radiation therapy can cause fatigue and localized side effects[6)]. Surgery carries risks of complications, including infections and changes in bowel function. Palliative care manages symptoms but doesn't cure the disease. Clinical trials offer experimental treatments, but they can

*Correspondence to: Yueran Wu, Chronic Disease Department, Jiujiang Center for Disease Control and Prevention, Jiujiang, 332000, CHINA
E-mail: yueranwu2@sina.com
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have unforeseen side effects, and not all participants receive active treatment. Treatment choices depend on individual factors and cancer stage[7]. Considering all these potential drawbacks of the current therapeutic approaches, stringent need to develop superior drug delivery technology has raised.

Natural phytochemicals found in plants offer several advantages over synthetic or semisynthetic molecules in treating colorectal cancer[8]. They tend to have lower toxicity, target multiple pathways, provide antioxidant and anti-inflammatory benefits, work synergistically, exhibit pharmacological diversity, and have chemo preventive potential. These attributes make them promising candidates for colorectal cancer treatment, offering a holistic and potentially safer approach to combat this complex disease[9].

Ampelopsis Radix, known as Bai Lian in traditional Chinese medicine, is derived from the roots of the Ampelopsis grossedentata plant, native to China and Asia. In traditional Chinese medicine, it's used for potential therapeutic benefits, including addressing conditions like diabetes, hypertension, and inflammation[10]. Active compounds in the root include stilbenes, flavonoids, and polyphenols, which may contribute to its medicinal properties. Research suggests possible anti-inflammatory, antioxidant, and anti-diabetic effects[11]. Ampelopsis Radix is typically prepared as a decoction or extract. While generally considered safe, its use should be guided by healthcare providers due to potential interactions or side effects. Ampelopsis Radix, sourced from Ampelopsis grossedentata roots, shows promise in potential cancer prevention and treatment. Studies indicate its antioxidant properties, safeguarding cells from cancer-related free radical damage[12]. Its anti-inflammatory compounds may reduce cancer risk associated with chronic inflammation. Additionally, Ampelopsis Radix may slow cancer cell growth (antiproliferative effects) and induce programmed cell death (apoptosis)[10]. It can support the immune system, aiding in cancer cell recognition and elimination. The herb may have chemo preventive potential, potentially impeding the initial stages of cancer development. While research is ongoing, these findings suggest Ampelopsis Radix's potential in cancer care. Very limited studies have been performed on Ampelopsis Radix with respect to colorectal cancer. Hence, there is urgent need to explore this plant in treatment of colorectal cancer.

Phytosomes are delivery systems that increase the absorption of plant-based phytochemicals. While not a direct cancer treatment, they can be part of cancer therapy by enhancing phytochemical effectiveness[13]. Phytosomes boost absorption, making phytochemicals more bioavailable, potentially as complementary therapies. This may enable lower, safer doses with reduced side effects. Phytosomes can be tailored for targeted drug delivery to minimize damage to healthy tissue, and they can enhance the synergistic effects of various phytochemicals, potentially strengthening their impact on cancer cells[14].

In light of the promising advantages associated with phytosomes, our research endeavours are focused on the development of phytosomes loaded with an ethanolic extract of Ampelopsis Radix for enhanced treatment of colorectal cancer. Notably, the utilization of phytosomes for the delivery of Ampelopsis Radix has yet to be explored, making our study a novel and pioneering investigation in this domain. Our aim is to harness the potential benefits of phytosomal-mediated drug delivery to optimize the therapeutic impact of Ampelopsis Radix in the context of colorectal cancer treatment.

2 Materials and Methods

2.1 Materials

Dried tuberous root of Ampelopsis Radix was purchased from local herbal drug company. Egg phosphatidylcholine (EPC) and Cholesterol were kindly gifted by Lipoid GmbH, Ludwigshafen Germany. Disodium hydrogen phosphate, potassium dihydrogen phosphate, sucrose and chloroform, Triton X 100, ethanol were purchased from Sigma Aldrich, USA. All other materials were of analytical or reagents grade.

2.2 Preparation of ethanolic extract of Ampelopsis Radix roots

The process involved transforming the dried tuberous root of Ampelopsis Radix into a fine powder using an electric grinding machine (Retsch; GM 300). This powder was then sieved through a 100-mesh screen to ensure uniform particle size and subsequently stored in airtight polythene bags. To extract the active constituents, 100 grams of the powdered drug were soaked in 500 milliliters of ethanol and subjected to a 6-hour extraction process in a Soxhlet extractor. The resulting mixture was then filtered through Whatman no. 1 filter paper. The ethanol extract was concentrated to reduce its volume using a rotary evaporator (Heidolph; Hei-VAP Core Standard G1 diagonal glassware Handlift) and a vacuum dryer (TVO-2 Vacuum Oven). Subsequently, this concentrated ethanolic extract was employed in the preparation of phytosomes[12].

2.3 Formulation of ethanolic extract of Ampelopsis Radix loaded phytosomes

The method of thin film hydration[9] has been used to formulate phytosomes. EPC and cholesterol with various molar concentrations (1:1, 1:2, 1:3 and 1:4) were dissolved in chloroform (5-10 mL) in the round bottom flask. Ethanolic extract 100 mg was added to lipidic solution. To evaporate the organic solvent under reduced pressure at 40℃, the rotary evaporator (Heidolph; Hei-VAP Core Standard G1 diagonal glassware Handlift) was used to achieve a very thin layer of dry lipids on the inner surface of the circular bottom flask. With a 15 mL saline phosphate buffer pH 7.4, this dry film was slowly hydrated. Mechanically, the resultant suspension was shaken for 1 h at room temperature using a shaker to shape the multilaminar phytosomes. The phytosomal dispersion was left at 40℃ overnight to ensure full hydration of the lipid. The extract-loaded phytosomes were isolated by centrifugation (Remi; C-30 Plus CPR-30 Plus) at 20,000 RPM for 3 hrs at −5℃ using ultracentrifuge cooling from the non-entrapped extract. For further research, suspension and powder were used. The prepared phytosomes were thus placed in a well-closed container and reconstituted with sterile water for injection for further use[15].

2.4 Characterisation of extract loaded phytosomes

2.4.1 Drug loading

To evaluate free ethanolic extract, which was isolated using the Sephadex G-50 mini axis, around 2.5 mL of phytosomal suspension was used. For 20 min, the suspension was centrifuged at 10000 RPM. For the digestion 0.1% v/v solution of Triton X-100 was used and elute was collected. It was analysed at 290 nm using UV spectrophotometer[16].

2.4.2 Surface morphology

Scanning electron microscopy (SEM) (Hitachi High-Tech's scanning electron microscopes SU3800) was used to check the surface morphology of the phytosomes. The working gap was held at 8.5–8.7 mm with an accelerating voltage of approximately 15.0 kV for zeta potential determination. By covering them with gold, the phytosomes must be made electrically conductive. Using adhesive tape with both-side adhesive, the gold-coated liposomes were securely mounted on a brass bowl. A vacuum (5 pa) in an ion sputter was preserved throughout the whole process[15].

2.4.3 Particle size, polydispersity index (PDI) and zeta potential

In double distilled water, the freeze-dried phytosomes were suspended and sonicated for 1 min prior to study. A Zetasizer (Malvern instruments; DTS Ver 4.10) at room temperature was used to count particle number, particle size distribution and zeta potential[17].

2.4.4 *In vitro* release study from phytosomes

The procedure of dialysis membrane diffusion was used to assess the release of the extract from the phytosomes. In short, an appropriately calculated quantity of pure ethanolic extract and phytosomal formulations filled in dialysis tubing (MW cut off 14000 Da) equal to 20 mg. The tubing was tied at both ends. In a buffer solution (pH 6.8, 200 mL, 37±2℃), this dialysis tube was suspended. The entire set up was mounted on magnetic stirrer and stirred at 150-rpm. The samples (5 mL) were collected every 1 h and a fresh 5 mL buffer was applied to preserve the sink state for 10 h and the spectrophotometrically release was estimated[15].

2.4.5 Cell line and cytotoxicity study

The cell line (HCT-116 and SW480 cell lines) acquired from a local hospital. Before treatment, cells were cultured for 24 hours in DMEM supplemented with 10% fetal bovine serum, 100 g/mL streptomycin, and 100 IU penicillin at 37℃ in a humidified environment containing 5% CO_2 at 1×10^4 cells/well in 96-well plates. For the period of 24 hours, the cell lines co-cultured with pure extract, and phytosomes at varied concentrations. The viability of the cells was studied by using MTT assay technique. MTT (1 mg/mL) was allowed to incubate with the substance under the investigation. The quantity of MTT transformed to insoluble formazan was used to confirm cell viability. The undissolved crystal was dissolved in a 1:14 (v/v) combination of 1M HCl and isopropyl alcohol and agitated for 20 minutes at room temperature[10].

2.4.6 Cellular uptake studies

This study was used to determine the accumulation of extract in (HCT-116 and SW480 cell lines) cell lines. In 12-well culture plates, HCT-116 and SW480 cell lines were seeded. Both cell lines were kept at a density of 10^4 cells per well and incubated for 20 hours. The test substances (pure extract, and phytosomes) were given to the grown cells at doses ranging from 0.02 to 2.0 M for 6 hours. The treated cells were trypsinized, washed thrice with physiological saline solution, centrifuged, and treated with lysis buffer before being sonicated. Using the HPLC technique, the accumulated extract was measured (in supernatants)[18].

2.4.7 *In vivo* anticancer activity of phytosomes

In vivo anticancer activity against colorectal cancer targeting assay was done in rats. The rats were divided into 3 groups containing 6 rats in each group. The rats were anesthetized with isoflurane and 1,2-dimethylhydrazine (DMH) injected intraperitoneally over period of 3 weeks to induce development of tumors in the distal colon. The animals were kept under observation until the colorectal cancer develops. The test substances (pure extract and extract loaded phytosomes) were administered intraperitoneally to rats at dose of. 2.5 mg/kg for 3 times at a gap of 3 days once. At each time point, mice were sacrificed and tumor was isolated. Antitumor activity was calculated by tumor volume, tumor weight. Ethanolic extract concentrations deposited in colon area were measured by scarifying rats at 1, 12 and 24 h post treatment. The colons were carefully removed and washed with PBS and soaked in 70% nitric acid for 12-15 h. The organs were digested at 90℃ for 2 h and then homogenized, centrifuged, supernatant was collected and analysed using HPLC method[19].

Table 1 Physicochemical parameters of PFD loaded liposomes.

Parameters	Ampelopsis Radix ethanolic extract loaded phytosomes					
Formulations	F1	F2	F3	F4	F5	F6
EPC:Cholesterol (mol)	1:0.5	1:1	1:1.5	1:2	1:2.5	1:3
Solution appearance	Turbid	Turbid	Turbid	Turbid	Hazy	Hazy
Particle size (nm)	85	105	115	121	126	130
PDI	0.25	0.125	0.35	0.27		
Zeta potential	12.26 ± 1.3	17.45 ± 1.1	15.11 ± 2.5	19.27 ± 3.5	16.90 ± 1.5	18.70 ± 2.5
% DLE	29.27	32.28	35.51	39.12	40.17	42.10

3 Results and Discussion

3.1 Physicochemical characterization of ethanolic extract loaded phytosomes

Film hydration techniques are essential for the development of phytosomes, advanced drug delivery systems that improve the bioavailability and therapeutic efficacy of plant-derived compounds, particularly phytoconstituents and herbal extracts. These techniques are crucial for several reasons[20]. Phytosomes, formed by complexing plant compounds with phospholipids using film hydration, create a lipid bilayer around the compounds. This bilayer enhances solubility in both water and lipid-based environments, boosting bioavailability and aiding the transport of hydrophobic phytochemicals through the gastrointestinal tract. Film hydration ensures the uniform and stable encapsulation of plant compounds, protecting them from degradation due to factors like oxidation, light, and moisture, thus extending the shelf life of phytosome products[21]. Furthermore, film hydration techniques allow for tailored, controlled release of phytosomes, supporting the maintenance of therapeutic concentrations over time. Phytosomes can also be designed for targeted drug delivery, improving pharmacokinetics and reducing the potential toxicity of certain plant compounds. Their versatility and compatibility with various dosage forms make them valuable tools for modern pharmaceutical and nutraceutical applications, ensuring safe and effective delivery of plant-derived therapeutics[22].

In our research, the film hydration technique has proven successful in the preparation of phytosomes loaded with Ampelopsis Radix extract. These phytosomes were created using varying molar concentrations of EPC and cholesterol, with ratios ranging from 1:0.5, 1:1, 1:1.5, 1:2, 1:2.5, to 1:3, as outlined in Table 1.

An increase in particle size was observed in the phytosomes as the EPC:Cholesterol ratio increased. The size expanded from 85 nm to 130 nm. Both EPC and cholesterol are essential constituents of the phytosomes lipid bilayer. Cholesterol's propensity to integrate into the phospholipid bilayer results in increased bilayer thickness as its concentration rises, leading to larger phytosomal particles. Secondly, cholesterol introduces a range of intermolecular interactions within the lipid bilayer[23]. This influences the packing of phospholipids, promoting the formation of more ordered and densely packed structures, thereby contributing to the observed increase in particle size. Furthermore, cholesterol's role in modulating bilayer fluidity is significant. Elevated cholesterol content reduces bilayer fluidity, rendering it more rigid. This increased rigidity results in the formation of larger, less deformable phytosomes[24]. Additionally, the larger particle size may be attributed to enhanced encapsulation of the active compound within the phytosomes structure, driven by changes in the EPC:Cholesterol ratio. Lastly, higher cholesterol content may lead to the aggregation of phytosomes, forming larger particles due to altered interactions between lipid molecules and encapsulated compounds. In conclusion, the augmentation in phytosomes size with an increasing EPC:Cholesterol ratio is a consequence of variations in lipid bilayer thickness, intermolecular interactions, bilayer rigidity, encapsulation efficiency, and potential aggregation[23]. The drug loading was also increased (29.27 to 42.10%) with increase in EPC:Cholesterol ratio. These factors collectively contribute to the observed size fluctuations in phytosomes. Moreover, it was noted that the zeta potential of the phytosomes ranged within an acceptable range. Maintaining this zeta potential is crucial for ensuring the stability of phytosomes formulations, as well as other formulations characterized by nano-sized dimensions.

3.2 SEM analysis

Our SEM analysis (Fig. 1) revealed that these phytosomes exhibit a spherical shape with a smooth surface. This morphology ensures a uniform and unbroken surface, a crucial feature in preventing ethanolic leakage from the phytosomes. It also serves to enhance the efficiency of encapsulation and loading, making this technique promising for the effective delivery of Ampelopsis Radix extract and other similar compounds.

3.3 *In vitro* release study from phytosomes

The *in vitro* release was studied from pure extract and phytosomal formulations in pH 6.8 phosphate buffer. Among all the formulations F6 batch showed excellent sus-

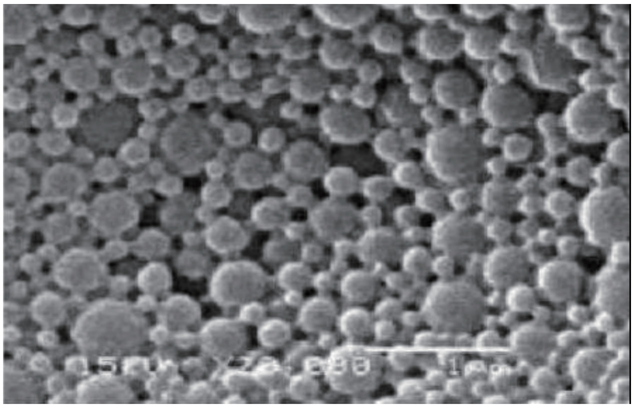

Fig. 1 SEM of Ampelopsis Radix ethanolic extract loaded phytosomes.

tained release behaviour in release media. The F6 phytosomal formulation demonstrated nearly 21% initial burst release followed by slow and gradual release up to 70 hours. As seen in Fig. 2, formulations containing lower ratios of EPC:cholesterol displayed relatively faster release profile as compared to F4 formulation ($p<0.05$). Phytosomes are a lipid-based drug delivery system used to enhance the bioavailability and stability of bioactive compounds like herbal extracts or pharmaceuticals. When creating phytosomes, the ratio of EPC to cholesterol is crucial for sustaining the release of the encapsulated extract. Phytosomes consist of lipid bilayers, similar to liposomes, and the structure of these bilayers is fundamental in controlling extract release[25]. EPC and cholesterol, the key components, influence the bilayer's properties. EPC increases bilayer fluidity, promoting extract release, while cholesterol reduces fluidity, making it less permeable. A higher EPC-to-cholesterol ratio creates a more fluid lipid bilayer, enabling sustained extract release over time. The extract can diffuse through the flexible bilayer, leading to a prolonged release. EPC contributes to phytosomes stability by preventing aggregation and extract leakage, which is crucial for maintaining controlled release[26]. The pure extract showed complete release within period of 1 hour. These observations clearly indicated that encapsulation of extract in lipidic vesicles sustained and retarded the release over the longer period of time which is quite essential in the treatment of colorectal cancer.

3.4 Cell line and cytotoxicity study

The *in vitro* cytotoxic assay was performed on human colorectal cancer cell lines.(HCT-116 and SW480) using MTT assay. In this assay pure extract and pure extract, and optimised phytosomal formulation (F6) were treated to both cell lines and incubated for 24 h. As can be seen from dose response curve (Fig. 3), both pure extract and F6 phytosomal formulations showed direct relation between concentration and cytotoxicity in HCT-116 and SW480 cell lines. Pure extract and F6 phytosomes at 0.25 and 0.5 μg/mL concentrations showed dose dependent cytotoxicity in HCT-116 and SW480 cell line. F6 phytosomes showed the maximum antiproliferative effect in this cell line as compare to that pure extract. The increased anticancer activity of F6 phytosomal formulation encapsulating an ethanolic extract of Ampelopsis Radix in colorectal cancer can be attributed to several factors. The F6 phytosomal formulation exhibited a significantly greater antiproliferative effect compared to the pure extract. This enhanced antiproliferative effect was primarily ascribed to improved cellular uptake. Initially, the phytosomes were taken up by endocytic cells, and subsequently, they escaped from these cells and reached the acidic lysosomes. Within the lysosomes, the drug was released gradually and in a sustained manner, continuously diffusing into the nuclear complex[27].

These lipid-based delivery systems enhance bioavailability, promoting higher concentrations of active compounds in the bloodstream and targeted delivery to cancer cells. Phytosomes offer sustained release, ensuring prolonged exposure to therapeutic agents, potentially boosting their efficacy against cancer cell growth[28]. Enhanced cellular uptake, facilitated by the lipid bilayer structure, aids in delivering the active compounds inside cancer cells. Syner-

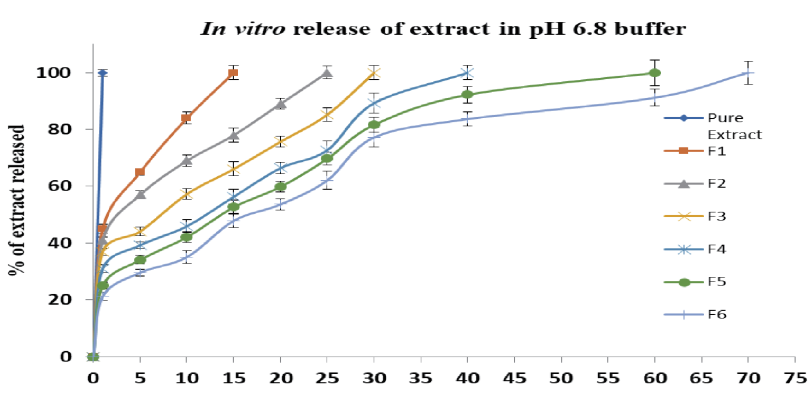

Fig. 2 *In vitro* release of extract in pH 6.8 phosphate buffer.

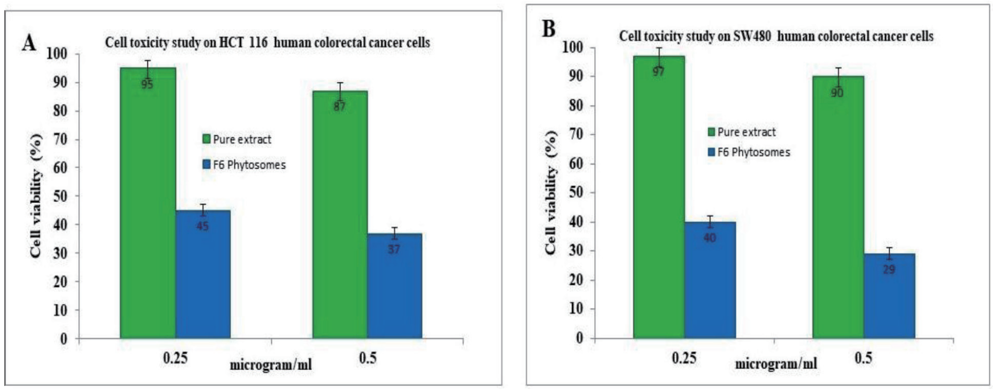

Fig. 3 Cell toxicity study on HCT-116 and SW480 human colorectal cancer cells.

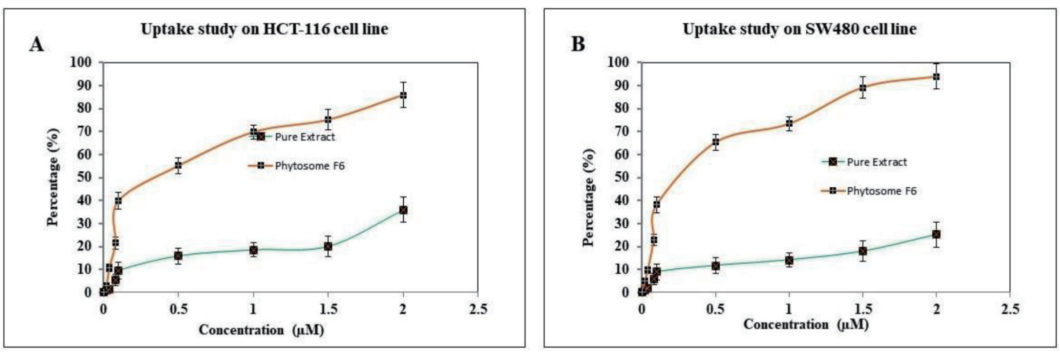

Fig. 4 Uptake study on HCT-116 and SW480 human colorectal cancer cells.

gistic effects may arise from combining Ampelopsis Radix bioactive compounds with phytosomal encapsulation, resulting in enhanced overall anticancer activity. Furthermore, the approach may minimize side effects by reducing exposure to healthy tissues[29].

3.5 Cellular uptake studies

The key prerequisite for the pharmacological activity of phytosomes is their successful internalization into cancerous tissues and cells, ensuring a sustained and continuous drug presence[25]. To assess this, a cellular uptake assay was conducted using two different cell lines, HCT-116 and SW480. These cells were exposed to both the pure extract and the phytosomal F6 formulation at concentrations ranging from 0.02 to 2.0 μM over an 8-hour period. The results revealed that the cellular uptake efficiencies of both the pure extract and the phytosomal F6 formulation were concentration-dependent for both cell lines, as illustrated in Fig. 4. However, the phytosomal F6 formulation exhibited significantly higher uptake efficiency compared to the pure extract. Specifically, in HCT-116 and SW480 cell lines, phytosomal F6 demonstrated 2.3-fold and 3.73-fold higher cellular uptake, respectively, compared to the pure extract. The enhanced uptake with phytosomal F6 was attributed to receptor-mediated endocytosis. In contrast, the pure extract showed notably lower cellular uptake due to non-specific adsorption or interaction with the cells[30]. Furthermore, as depicted in Fig. 4, the cellular uptake efficiencies of both the pure extract and phytosomal F6 formulation were found to be time-dependent, reaching saturation after an 8-hour incubation period.

3.6 *In vivo* anticancer activity of phytosomes

In the *in vivo* assessment of antitumor efficacy, both the pure extract and the phytosomal F6 formulation were tested in a rat model with distal colon cancer induced by DMH. As depicted in Fig. 5, the untreated control group exhibited a tumor volume of 820 mm^3. In contrast, rats treated with the pure extract and the phytosomal F6 formulation displayed significantly suppressed tumor growth, with tumor volumes of 820 mm^3 and 49 mm^3, respectively. The highest antitumor efficacy was observed in animals treated with the phytosomal F6 formulation. Notably, this formulation maintained a consistently small tumor volume throughout the study, with nearly 90% tumor regression, underscoring its effectiveness in controlling cancer metastasis. In terms of tumor weight (Fig. 5B), the phytosomal F6 formulation yielded the smallest tumors (0.35 g), followed by the pure extract (0.95 g), while the control group had the largest tumor size (1.25 g). The superior antitumor activity of the phytosomal F6 formulation was attributed to its enhanced cellular uptake by cancerous cells[30]. Addi-

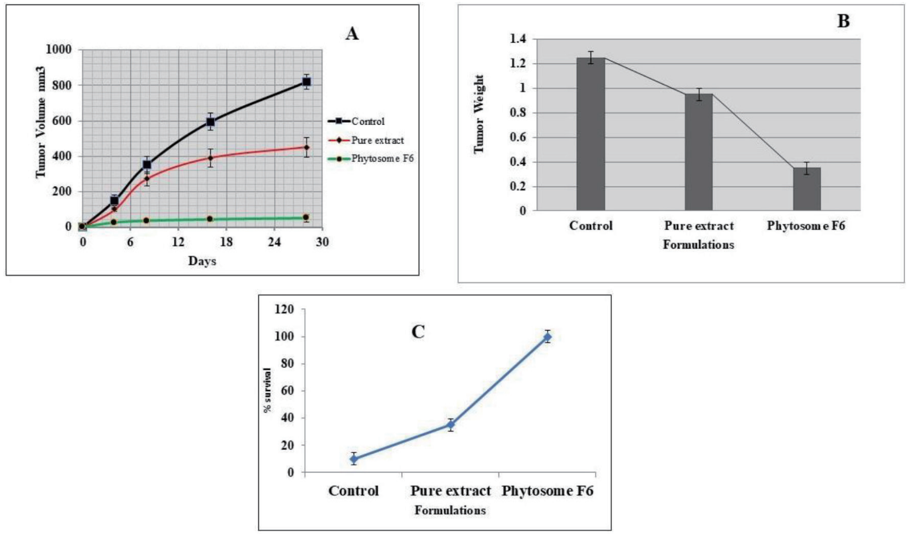

Fig. 5 Anticancer activity of phytosomes and its effect on tumor volume (A) tumor weight (B) and survival rate (C).

tionally, the median survival time for the phytosomal F6 formulation group reached 100%, while the untreated group exhibited the lowest survival rate, with only 10% at the study's conclusion, as shown in Fig. 5C.

4 Conclusion

In conclusion, the development and evaluation of Ampelopsis Radix ethanolic extract loaded phytosomes have demonstrated significant promise for improved efficacy in the treatment of colorectal cancer. The phytosomes exhibited favorable characteristics, including spherical shape, appropriate particle size, zeta potential, and high encapsulation efficiency, ensuring effective drug delivery. *In vitro* studies revealed the phytosomes' ability to release the drug in a manner suitable for cancer treatment, resulting in maximum antiproliferative activity, high cellular uptake, and superior cytotoxicity against HCT-116 and SW480 cell lines. Importantly, *in vivo* antitumor activity studies demonstrated a remarkable 100% survival rate in the phytosomes-treated group, affirming their safety and efficacy. Overall, this research underscores the substantial anticancer potential of Ampelopsis Radix ethanolic extract loaded phytosomes, supporting their potential utility for enhanced efficacy in the management of colorectal cancer.

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Author's Contributions

All authors contributed equally in this research work with respect to Conceptualization and methodology, Data collection, Data analysis and interpretation, Drafting the article.

Conflict of Interest

Authors do not report any conflict of interest.

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